# CONTINUOUS IMAGES OF CONNECTED SUBSPACES ARE CONNECTED

THE STACKS PROJECT AUTHORS

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*Allowed standard prerequisites (editorial).*

- The definition of a connected subset through the absence of a nontrivial relatively open and closed subset, and the continuity of the map. Set up in the author's section on connected components, immediately before the counted statement. Fixes the hypothesis on the subset and the conclusion about its image.

**Lemma 0.1.** *Let $f : X \to Y$ be a continuous map of topological spaces. If $E \subset X$ is a connected subset, then $f(E) \subset Y$ is connected as well.*

*Proof.* Let $A \subset f(E)$ an open and closed subset of $f(E)$. Because $f$ is continuous, $f^{-1}(A)$ is an open and closed subset of $E$. As $E$ is connected, $f^{-1}(A) = \emptyset$ or $f^{-1}(A) = E$. However, $A \subset f(E)$ implies that $A = f(f^{-1}(A))$. Indeed, if $x \in f(f^{-1}(A))$ then there is $y \in f^{-1}(A)$ such that $f(y) = x$ and because $y \in f^{-1}(A)$, we have $f(y) \in A$ i.e. $x \in A$. Reciprocally, if $x \in A$, $A \subset f(E)$ implies that there is $y \in E$ such that $f(y) = x$. Therefore $y \in f^{-1}(A)$, and then $x \in f(f^{-1}(A))$. Thus $A = \emptyset$ or $A = f(E)$ proving that $f(E)$ is connected. □

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